CATEGORICAL SYLLOGISMS

Rules and Determining Fallacies

The Rules of Categorical Syllogism

The rules are divided into two:

Rules 1-4

Rules governing the **terms**

Rules 5-8

Rules governing the **propositions**

The RULES are as follows:

1. The middle term must always be taken in the same sense.
2. The major term (P) and minor term (S) cannot have a greater extension in the conclusion than in the premise.
3. The middle term (M) should not occur in the conclusion.
4. The middle term (M) must be distributed universally, at least once, in the premises.
5. Two affirmative premises cannot give a negative conclusion.
6. From two negatives, nothing follows.
7. From two particular premises, nothing follows.
8. The conclusion follows the weaker premise.

The FALLACIES are as follows:

1. Fallacy of Equivocation
2. Fallcies of Illicit Minor and Illicit Major
3. Fallacy of Misplaced Middle Term
4. Fallacy of Undistributed Middle Term
5. Fallacy of a Negative Conclusion Drawn from Affirmative Premises
6. Fallacy of Exclusive/Negative Premises
7. Fallacy of Two Particular Premises
8. Fallacy of the Universal Conclusion Drawn from a Particular Premise

BE FAMILIAR WITH THE FOLLOWING:

M	Middle term (universal)
P	Major term (universal)
S	Minor term (universal)
p	major term (particular)
s	minor term (particular)
+	affirmative proposition
-	negative proposition

RULE 1: The **middle term** must always be taken in the same sense.

FALLACY OF EQUIVOCATION

A small hand-held device dragged to move a cursor on a computer screen is a small rodent that has relatively large ears and eyes and a long tail.





! EQUIVOCATION

An argument that uses one word to mean two different things

RULE 1: The **middle term** must always be taken in the same sense.

FALLACY OF EQUIVOCATION

Some handheld devices are mice.
A rodent is a mice.
Hence, a handheld device is a rodent.





RULE 2: The **major term (P)** and minor **term (S)** cannot have a greater extension in the conclusion than in the premise.

ILLICIT MAJOR

The predicate is **universal in the conclusion**, not in the major premise

ILLICIT MINOR

The subject is **universal in the conclusion**, not in the minor premise

All hammers are tools; **M + P**

But, no chisels are hammers; **S - M**

:. No chisels are tools. **S - P**

FALLACY OF ILLICIT MAJOR



The effect (conclusion) becomes **greater than the cause** (premises), which is **impossible**.

RULE 3: The **middle term** should not occur in the conclusion.

FALLACY OF MISPLACED MIDDLE TERM

A steward is a flight attendant; **M + p**

But, a steward is a male; **M + s**

:. A steward is a male flight attendant.

M + sp

RULE 4: The **middle term** must be distributed universally, at least once, in the premises

FALLACY OF UNDISTRIBUTED MIDDLE TERM



What is true of its whole is true of its parts.

All stewardesses are females. P + M

But, all mothers are females. S + M

:. All mothers are stewardesses. S + P

All apples are fruits. P + M

But, some fruits are mangoes. m + s

:. Some mangoes are apples. s + p

RULE 5: Two affirmative premises **cannot** give a negative conclusion

FALLACY OF A NEGATIVE CONCLUSION DRAWN FROM AFFIRMATIVE PREMISES



A negative conclusion would express **something that is not contained in the premises**.

All stones are hard; +

Some diamonds are stone; +

:. Some diamonds are not hard. -

RULE 6: From two negatives, nothing follows.

FALLACY OF EXCLUSIVE/NEGATIVE PREMISES



No relationship can be established ☹

No conclusion can be inferred.

A chair is not a table;

But, a table is not a pen;

:. A pen is not a chair.

RULE 7: From two particular premises, nothing follows.

FALLACY OF TWO PARTICULAR PREMISES

a

If both particular premises are **affirmative**, then the subjects and predicates are **particular** (violates rule #4)

b

If both particular premises are **negative**, then the syllogism violates rule #6.

c

If one particular premise is **affirmative**, and the other **negative**, then the syllogisms are fallacious.

Some cats are pets; **m + p**

But, some pythons are not cats; **s – m**

:. Some pythons are not pets. **s – p**

Some horseback riders are not males; **m - p**

But, some drivers are horseback riders; **s + m**

:. Some drivers are not males. **s – p**

RULE 8: The conclusion follows the weaker premise.

FALLACY OF THE UNIVERSAL CONCLUSION DRAWN FROM A PARTICULAR PREMISE



If one premise is **universal**, and the other is **particular**, the conclusion should be **particular.**

All mongoloids are mentally retarded; M + P

But, some mongoloids are special children; m + s

:. All special children are mentally retarded. S + P

DRILLS

Identify the fallacy(ies) committed.

Rape is immoral;
But, cheating is immoral;
:. Cheating is rape.

A martyr dies for a noble cause;

But, some wives are martyrs;

:. Some wives do not die for a noble cause.

Waiters wait;

But, true love waits;

:. True love is a waiter.

Horror stories send chilling effects;

But, murder stories are horror stories;

:. Murder stories don’t send chilling effects.

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